Asma Guenaien

Software Engineer | Full-Stack Developer (MERN, Symfony, Angular) · Secure Web Applications

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Professional Summary

Computer Science graduate specializing in Intelligent Systems, with a background in research and full-stack web development. Passionate about building innovative, secure, and user-centered applications with React, TypeScript, Node.js, and modern web technologies. My thesis delivered a secure web platform that reduced re-identification risk on medical data from 58% to 10%. Interested in software development, artificial intelligence, and cybersecurity, I am eager to contribute to impactful projects and continue growing professionally.

Experience

Research Master's Thesis – Secure Web Platform for Health Data Privacy Feb 2026 – Oct 2026
ENSIT (National Higher School of Engineers of Tunis) *Tunis, Tunisia*

- Designed and developed a web-based platform to assess and mitigate privacy risks in medical datasets, from architecture to implementation.
- Implemented and evaluated anonymization techniques with quantitative privacy metrics, **reducing re-identification risk from 58% to 10%**.
- Integrated a Retrieval-Augmented Generation (RAG) module giving users context-aware access to research knowledge while preserving data confidentiality.
- Conducted a literature review on medical data privacy and re-identification risks to define the technical approach.
- Defended the thesis in October 2026 with **Highest Distinction**.
- *Skills: Full-Stack Development, Data Anonymization, RAG Integration, Secure Design*

End-of-Study Project Intern – Full-Stack Developer Jan 2024 – May 2024
Progress Engineering *Tunis, Tunisia*

- Led a company-wide hardware management web application (MERN stack) from design to deployment, allowing employees to submit repair requests for laptops, scanners, and phones.
- Streamlined internal hardware management and request handling.

Software Developer Intern Jul 2023 – Aug 2023
Progress Engineering *Tunis, Tunisia*

- Built a Symfony web application for submitting and tracking vacation requests, improving HR efficiency.
- Worked with the IT team to ensure smooth integration with existing systems.

Education

Master's Degree, Computer Science – Intelligent Systems (Research) Sept 2024 – Oct 2026
ENSIT (National Higher School of Engineers of Tunis) *Tunis, Tunisia*

- Thesis: *Privacy Protection for Patient-Generated Health Data in the Internet of Medical Things (IoMT).* Research on multilayer security architecture, re-identification risk assessment, k-anonymity-based anonymization, and security recommendations through a RAG module.
- Thesis defense: October 2026, Highest Distinction. Focus: IoT, Artificial Intelligence, Automation.

SOC Analyst Cybersecurity Bootcamp (FTD Scholarship) Jul 2024 – Oct 2024
TED University

- CCNA, CyberOps Associate, Certified SOC Analyst (CSA), Certified Ethical Hacker (CEH).
- Focus: network security, incident response, ethical hacking, hands-on cybersecurity tools.

Bachelor's Degree, Computer Science – Software Engineering & Information Systems Sept 2021 – Jun 2024
Faculty of Sciences of Bizerte *Bizerte, Tunisia*

- Relevant courses: Software Engineering, Information Systems, Web Development, Artificial Intelligence, Databases.

Selected Projects

- **Secure Health Data Platform (Thesis):** full-stack web platform to measure and reduce re-identification risk in medical datasets; risk reduced from 58% to 10%.
- **Hardware Management App (MERN):** repair-request workflow for company devices, delivered end-to-end (MongoDB, Express.js, React, Node.js).
- **Vacation Request App (Symfony):** leave submission and approval workflow for HR.
- **E-commerce Platform (PHP/Laravel):** product catalog, shopping cart, secure checkout, and payment API integration.
- **University Portal Database (MySQL):** student registration, course enrollment, and grade tracking.

Technical Skills

- **Development:** JavaScript, TypeScript, Python, Java, HTML/CSS, React, Node.js, Express.js, MERN stack, Symfony (PHP), Angular, Laravel, Odoo
- **Databases & Systems:** MySQL, PostgreSQL, MongoDB, Linux system administration
- **Security:** secure application design, vulnerability management, threat hunting, penetration testing; Metasploit, Nmap, Wireshark, Burp Suite
- **Data Privacy:** anonymization (k-anonymity), re-identification risk assessment, RAG integration

Certifications

- Certified Ethical Hacker (CEH) – in progress
- Foundations of Networks, LinkedIn Learning (2022)
- Object-Oriented Programming in Java, Udemy (2023)
- The Essential Basics of the Symfony PHP Framework, Udemy (2023)
- Programming Foundations: Fundamentals, Coursera (2022)

Leadership & Activities

- General Secretary, Google Club ISIGK (Sept 2022 – Jun 2023)
- Member, JCI Radès (Sept 2023 – present) · Securinet ISI (Sept 2024 – present) · AIESEC Bizerte (Oct 2024 – present)

Languages

Arabic (native) · French (full professional) · English (professional working) · German (limited working) · Italian and Turkish (elementary)